

Analyzing longitudinal data using GLMMs

1 Exercise 1

We consider here data coming from studies conducted in the Northern Territory of Australia. In these studies, medical doctors compared different ways of treating the bacterium *Haemophilus influenzae*, often found on children suffering from ear infections. The data set, which is called `bacteria` by default, is available in the **MASS** R package and can be loaded through `data(bacteria)`.

The data features 50 children with diagnosis of ear infection. They were randomized to receive either a drug, a placebo or a drug together with active encouragements to comply with taking the drug. There are therefore three treatment groups which are the three levels of the variable `trt`: `placebo`, `drug`, and `drug+` (i.e. with encouragements). The presence of *H.influenzae* was checked at weeks 0, 2, 4, 6 and 11. Thirty of the checks were missing and are not included in the data set which consequently has only 220 observations (instead of 250). Description of all variables is available by executing `help(bacteria)`.

- a) Suggest several models and fit them to the data in order to establish whether the drug is effective against the bacterium or not.
- b) Does encouraging the child increases the drug effect significantly?
- c) Does a GLMM fits the data better than a GLM? How can you explain this?

2 Exercise 2

Seizures are often observed on epileptic children and are believed to be caused by abnormal brain functioning. The meaning of “abnormal” is yet to be fully understood, so far medications are mainly developed to reduce symptoms.

The `seizure` data set is introduced on slide 214 of the course and contains 4 variables:

- `y`, the number of seizures,
- `Time`, the time period in which seizures were encountered,

- `Subject`, the subject identification,
- and the treatment allocation variable `group`, taking values `placebo` or `progabide`.

Fit a GLMM to these data in order to determine whether the progabide treatment is effective at reducing the rate of seizures among epileptic patients. Compare the GLMM fit with the GEE fit presented in class.